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RESEARCH ARTICLE

Hosting-Capacity Improvement Using Rooftop Photovoltaics and Decentralized BESS in Low Voltage Grids

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ABSTRACT Voltage regulation in low-voltage (LV) distribution grids is essential to increase the penetration of photovoltaic (PV) rooftop units. Hence, PV inverters are commonly embedded with volt-var (VV) and volt-watt (VW) control functions to support the grid voltage regulation within the grid codes. Such an approach usually increases the LV grid hosting capacity (HC) by 1.5 to 2 times at the cost of energy curtail. Thus, the use of decentralized battery energy storage systems (BESS) near to nodes sensible to overvoltage become appealing, as it stores energy without creating current circulation within the grid. However, there are still gaps in sizing of decentralized BESSs and in quantifying their improvement in terms of HC. Thus, this paper contributes with a comprehensive study that evaluates the HC of a generic LV grid comprised of PV units with embedded VV and VW control functions, and proposes a methodology to guide the sizing of small decentralized BESS used to increase the grid HC. Using OpenDSS software, the VV and VW control functions are evaluated and compared with a base scenario where PV units operate at a unity power factor. The active power curtailment is used to size decentralized BESS to absorb excess energy, avoiding PV active power derating. Results show that BESS units, sized at about 5-10% of the PV power capacity, can reduce revenue impacts and increase HC by over 30% in LV grids.

INDEX TERMS Hosting capacity, volt-var, volt-watt, photovoltaic, battery energy storage system.

I. INTRODUCTION

The large-scale connection of distributed energy resources (DERs) to electrical grids is overwhelming power systems, which in many cases cannot handle the excess power injected by end-users. The hosting capacity (HC) metric is the maximum amount of power a grid can accommodate relative to the upstream transformer's rated power, without compromising power quality (PQ) indices.

The voltage regulation performed by photovoltaic (PV) systems by means of volt-var (VV) and volt-watt (VW) control functions, although it improves the HC value of the

low-voltage (LV) grid by 1.5 to 2 times [1], results in curtailed energy from the PV units. Thus, the use of decentralized battery energy storage systems (BESSs) near the nodes of the grid most sensitive to overvoltage becomes an appealing solution, as it stores the energy that would otherwise be discarded without creating current circulation into the electrical grid. However, this solution still lacks studies to quantify its improvement in the HC of the LV grid, along with methods for sizing the BESSs and assessing economic feasibility. This paper aims to address the first two gaps.

Integrating BESS into LV grids offers numerous benefits [2]. However, several practical deployment challenges must be addressed to fully realize these advantages, including grid compatibility and infrastructure limita-

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tions [3], regulatory and market barriers [4], optimal sizing and placement [5], and consumer engagement and awareness [6].

Recent studies highlight the latest advancements in BESS technology and its integration into LV grids. In [7], a comprehensive review of energy storage systems is presented, including their applications for grid integration and the types of storage technologies. Similarly, in [8], the role of BESS as both a load and a generator is detailed, demonstrating its capability to provide voltage support and enhance the stability of distribution grids. The authors of [9] explore how BESS can store excess energy from DERs during high-generation periods and discharge it during peak demand, thereby enhancing grid stability. In [10], a comprehensive overview of the current state and future trends in grid-connected energy storage, highlighting the challenges and opportunities associated with integrating these systems into modern power grid. Reference [11] provides an overview of the critical role of energy storage systems in enhancing the flexibility, reliability, and stability of modern power systems, especially with the increasing penetration of DERs. Finally, [12] serves as a valuable resource for understanding the current state of BESS technologies, their integration into power systems, and the key challenges that must be addressed to enhance their effectiveness and broader adoption.

The technical literature highlights several noteworthy real-world BESS projects worldwide, which are connected to medium-voltage (MV) distribution systems. References [13] and [14] examine several large-scale BESS projects, offering insights based on operational experiences, summarized in Table 1. Furthermore, [15] examines the decommissioning of the Tehachapi Energy Storage Project (TSP). Commissioned in 2014 and decommissioned in 2022, the TSP (8 MW/32 MWh, USA) was one of the largest lithium-ion battery storage systems in North America, providing valuable insights into its long-term operations and maintenance. In [16], the UE LV Grid BESS in Australia is presented, involving the deployment of 40 custom-built batteries of 30 kW/60 kWh batteries mounted to electricity poles across United Energy's distribution, providing up to two hours of stored energy to support more rooftop solar and manage peak demand across the LV distribution grid.

A study of 50,000 LV grids revealed that 85% reached saturation when DER penetration exceeded 20% of the medium-voltage/low-voltage (MV/LV) transformer's rated power [17]. The most affected PQ indices under high PV penetration are overvoltage, conductor thermal stress, and voltage imbalances, although other issues may also arise [18]. Another reference assesses the HC of LV grids with PV penetration in two regions of Sweden [19], using deterministic and stochastic methods to evaluate the probability of PV installations and their effects. This analysis aids in investment planning and decision-making. From a different perspective, the authors of [20] examine the adverse effects of high PV penetration on distribution grids, focusing on grid protection and the impact of PQ issues on system operation.

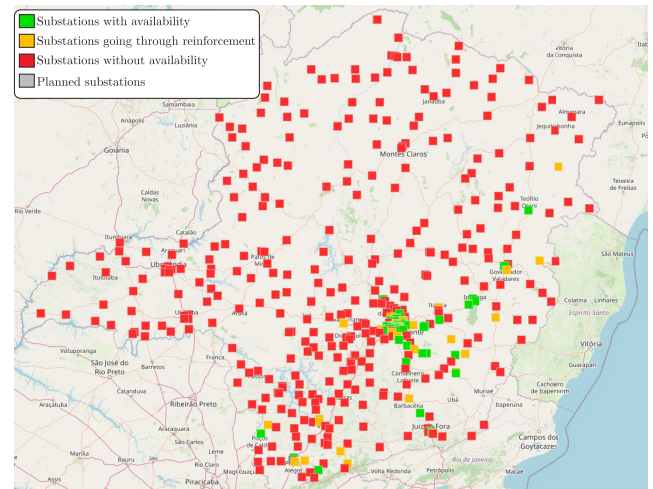


FIGURE 1. Distributed generation availability map - Minas Gerais, Brazil.

The authors of [1] propose methods to increase PV-based DER connections without surpassing the grid HC. Similarly, [21] shows a two-part review on the effects of PV penetration on LV grids, detailing common issues such as voltage fluctuations and current rises, and how these issues limit new PV connections to maintain grid stability. Therefore standards have been developed to enumerate requirements for DER interconnection with electrical power systems, focusing on performance, testing, safety, and responses to abnormal conditions [22]. For instance, in [22], DER operations are categorized into three functional groups. The first group addresses DER reactive power capability and voltage regulation, with two subcategories. Category A includes inverters operating at a unity power factor, supporting basic voltage regulation functions such as constant power factor, constant reactive power, and/or VV control mode. Category B extends these DER capabilities, adding VW control mode to the functions of category A. The other two DER groups, II and III, are designated for abnormal conditions, with more restrictive disturbance ride-through requirements. In [23], the state-of-the-art of LVRT (low-voltage ride-through) techniques are presented, focusing on synchronization methods and current injection techniques based on PV-DER, with an emphasis on the X/R ratio.

Fig. 1 shows the status of electrical substations in Minas Gerais state, Brazil, to host DERs in October/2024. Red squares are substations saturated (80.4%). Orange squares are substations with planned upgrades (6.7%). Green squares are substations available to host PV units (12.7%). Gray squares are scheduled for installation but are not yet operational (0.23%) [24]. To enhance HC values, grid reinforcements that commonly involve increasing conductor sizes or upgrading to higher-rated transformers. On-load tap changers are another option but can lead to higher maintenance costs and increased failure rates due to frequent tap changes from rapid voltage fluctuations caused by high PV penetration. These

TABLE 1. Summary of worldwide BESS projects connected to MV distribution systems and their main functionalities.

Project	Location	Capacity	Main Functionalities
Monterey Vistra Moss Landing	USA	1500 MW / 6000 MWh	Grid reliability, DER integration, peak shaving
GS Yuasa-Kita Toyotomi	Japan	240 MW / 720 MWh	Grid stability, DER integration, power smooth
Neurath & Hamm RWE	Germany	220 MW / 235 MWh	Peak shaving, DER support, frequency regulation
Hornsedale Power Reserve	Australia	150 MW / 194 MWh	Grid stability, DER integration, frequency regulation
Carbon Free plant	Korea	100 MW / 200 MWh	Grid stability and reability, DER integration, frequency regulation

solutions require substantial investment, long-term planning, and temporary outages for installation, as discussed in [25].

The authors of [25] also explore emerging techniques for improving grid HC values: (i) thyristor-controlled reactor (TCR), which promptly compensates for reactive power; (ii) the static var compensator (SVC), which regulates reactive power to enhance system operation; and (iii) the static synchronous compensator (STATCOM), a shunt synchronous voltage generator that exchanges reactive power with the distribution system. While these methods effectively manage reactive power in grids with high X/R ratios, they are less effective in low X/R ratio LV grids, which require solutions that perform well in both scenarios and do not increase current circulation within the electric power system.

As DERs are integrated into the grid, they can be concomitantly used for grid-supportive functions. Voltage control strategies can be embedded in PV and BESS inverters to enhance grid reliability and HC. Although promising, this approach requires further alignment with existing standards and grid codes, as discussed in [21]. DER inverter control strategies provide a effective solution for minimizing PQ concerns in LV grids. They are adaptable, managing reactive power (i.e., VV control) for grids with high X/R ratios and active power curtailments (i.e., VW control) for grids with low X/R ratios. However, VV control can over-stress feeders, while VW control may incur financial penalties for end-users.

Under such a context, [26] shows how overvoltage limits PV integration into LV grids. The study evaluates various DER techniques, including VV control and PF(P)-power factor by injected active power, and shows that these methods can increase average PV penetration by 38% and improve HC values from 36% to 50%. Moreover, the authors of [27] verified that inverters located farther from the point of common coupling (PCC) typically activate control techniques earlier than those closer to the PCC. This allows inverters near the PCC to operate at full capacity while those farther away regulate the voltage profile at the grid's end. This is overcome in [28] by equitable active power curtailment.

In [29], or VV and VW control are assessed using a modified IEEE 13-bus grid, focusing on the prioritization of reactive power versus active power while respecting DER power ratings. The authors found that VV control yielded the best results in improving HC, being approximately twice as effective as VW control, and fixed power factor in enhancing voltage regulation. The authors of [30] compared fixed active power curtailment to VW control for regulating voltage fluctuations. The fixed approach reduced PV active power

based on its rated capacity, setting a minimum injection threshold of 70%, which resulted in a total power reduction of 6%. In contrast, the VW control function, which relies on local voltage measurements, achieved 1% reduction under the same conditions. This concludes that 5% of the annual energy yield was unnecessarily curtailed in such a study. The authors of [31] confirmed that active power curtailment based on VW control effectively mitigates technical issues in LV grids with high PV penetration.

Several studies have explored the potential of BESS to enhance grid HC in power systems. For instance, the authors of [32] compares three energy storage configurations: a large battery at the substation, medium-sized community storage, and smaller individual units. The results revealed that community storage is the most cost-effective option for increasing HC. While community storage achieves a slightly lower HC increase (35.4%) compared to individual units (37.5%), it is more economical due to reduced installation costs. Nonetheless, reaching 100% PV penetration still requires grid reinforcements. In [33], high PV penetration with storage operations is investigated using plug-in electric vehicle batteries. The study highlighted that small vehicle storage units could support the grid by storing curtailed energy. Proper usage of these units occurred around 11:30 a.m., which coincides with peak PV generation, and there is about a 50% chance of a vehicle being parked during this period [34]. Table 2 summarizes papers that explore strategies to address grid HC issues, and maps each paper to its respective approach.

In [35], a voltage sensitivity-related hybrid coordination algorithm is proposed for application in active distribution grids with the integration of PV and BESS. In this algorithm, VV control is given higher priority than VW control. The authors of [36] propose an event-triggered hybrid voltage regulation method. They introduce an event-triggered communication strategy based on voltage changes and sensitivity. A coordinated RPC-APC (reactive power compensation–active power compensation) approach is applied for voltage violations, while a local RPC-APC strategy is used to enable rapid voltage suppression without communication. Additionally, the minimum required BESS sizing is determined based on the total APC needs. The authors of [37] conducted studies with BESS equipped with VV and VW controls in grids with various X/R ratios, including both inductive and resistive grids. A sequence of local actions (by the BESS) and centralized actions (via a microgrid controller) using VV and VW controls is

TABLE 2. Comparison between the proposed paper and adherent state-of-the-art papers.

Refs	HC value	Reactive power control	Active power curtail	Decent. BESS units	BESS sizing
[21]	✓	✓	✓		
[25]	✓	✓	✓		
[27]	✓	✓			
[29]	✓	✓	✓		
[31]	✓		✓		
[32]	✓			✓	
[33]	✓			✓	
[38]	✓	✓		✓	
[39], [40]	✓				
[41], [42]	✓	✓			
Herein	✓	✓	✓	✓	✓

performed to mitigate overvoltage at the grid active nodes. VV control was more effective in highly inductive grids ($X/R = 10$), reducing the number of overvoltage nodes. In resistive grids ($X/R = 1$), VV had limited effectiveness, requiring VW control to address the overvoltage. However, the sizing of the BESS and the grid HC were not assessed.

This paper evaluates the HC value of a resistive LV grid and assesses the benefits of various autonomous voltage control strategies, including VV control, VW control, and a combination of both, along with decentralized BESS. The main contributions of this paper are:

- a method for sizing small decentralized BESS to operate alongside rooftop PV units in LV grids, that increases the grid HC and reduces the amount of power wasted due to PV active power curtailment control;
- analysis and demonstration of the advantages of using decentralized BESS in combination with PV generators in LV grids;
- HC estimation tool based on predefined premises for LV grids.

The results are evaluated and compared using a validated base scenario implemented in OpenDSS software and assessed against figures of merit to quantify the HC gains achieved with the proposed tool.

This paper is structured as follows: Section II covers the relevant definitions and premises used in this study. Section III details the proposed tool for HC estimation and sizing of decentralized BESS. Section IV describes the case study grid and its implementation in OpenDSS. Section V shows the simulation results, while Section VI concludes the paper.

II. DEFINITIONS AND PREMISES

A. HOSTING CAPACITY

Grid HC is a statistical metric that quantifies the maximum DER power that can be integrated into the grid before PQ metrics are violated. To estimate the HC value, four steps are required: i) identify the performance indicators most sensitive to high DER penetration; ii) establish the operational limits of indicators; iii) conduct a power flow analysis; and iv) set the HC value based on statistics. The HC is computed using (1),

which divides the total PV active power by the rated power of the upstream interfacing transformer.

$$HC(\%) = 100 \times \frac{\sum_{n=1}^N P_{pv,n}}{T_{nom}} \quad (1)$$

where $HC(\%)$ is the HC as a percentage, N is the total number of PVs, $P_{pv,n}$ is the active power inject into the grid by the n -th PV unit, and T_{nom} is the rated power of the MV-LV transformer. Additionally, the authors of [43] emphasize the uncertainty in the HC identification process, recommending that results should be analyzed as a data set rather than a single value. This sensitivity arises from the impact of DER installation locations, variations in DER ratings, power grid infrastructure, and the intermittent nature of the power sources on HC values.

B. PREMISES

The HC value relies on national regulations that set acceptable operating ranges, such as voltage levels, permissible current magnitudes, and frequency deviations. In this paper, the operational limits are set by PRODIST's Module 8 [44], which addresses quality metrics in Brazilian electrical distribution grids. This study focuses on three main issues in LV grids:

- **Voltage variation:** There are three types of supply conditions: normal, precarious, and critical. Table 3 outlines the voltage measurement ranges for each condition. Herein, a voltage violation is defined as the grid operating in precarious conditions for more than 5% of the simulation period.
- **Current violations:** The operational limits of each device are compared with the measured current to assess compliance. A current violation is defined when the current through an element exceeds its thermal limit for more than 5% of the simulation period.
- **Voltage imbalance:** The allowable voltage imbalance is 3% for LV grids. This study calculates voltage imbalance using positive and negative sequence voltages, as defined in (2). A voltage imbalance violation occurs when the imbalance exceeds 3% for more than 5% of the simulation period.

$$UF(\%) = \frac{V_2}{V_1} \times 100 \quad (2)$$

where $UF(\%)$ is the voltage unbalance factor, and V_1 and V_2 are the voltage positive and negative sequences, respectively.

1) VOLTAGE CONTROL STRATEGIES

A comprehensive report shown in [45] overview the functionality embedded in the smart inverters. Additionally, [22] is the main standard for the interconnection and interoperability of DERs with the grid, regulating DER operations and specifying requirements to improve grid performance. Among the functionalities described in [22] and [45], the voltage control technique monitors the local

TABLE 3. Description of the voltage operation ranges.

Operation condition	Voltage range description
Normal supply	Values between 0.92 pu and 1.05 pu
Precarious supply	Undervoltage: values between 0.86 pu and 0.92 pu Overvoltage: values between 1.05 pu and 1.06 pu
Critical supply	Undervoltage: values smaller than 0.86 pu Overvoltage: values greater than 1.06 pu

voltage of DER and adjusts its output power to maintain performance standards as outlined in Section II-B. This method can be effective, but may have potential drawbacks such as the increase in current level due to high reactive power circulation. Moreover, DER's owner might experience reduced revenue, because VW control may limit the amount of energy produced by PV units.

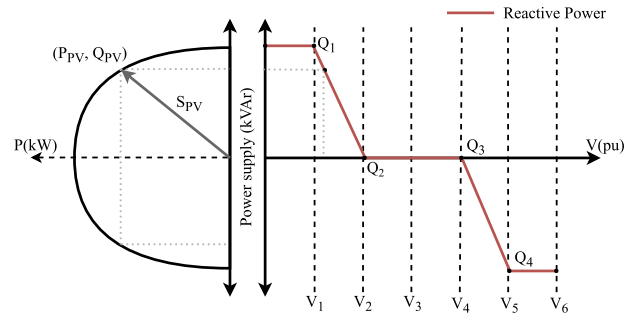
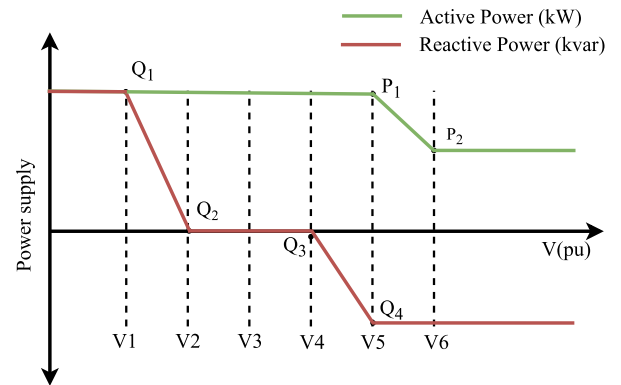
The primary goal of the VV control is maintain acceptable voltage profiles by regulating the output reactive power of a PV unit. If the voltage exceeds the upper limit, the system absorbs reactive power, operating in the inductive zone. Conversely, if the voltage falls below the lower limit, the PV injects reactive power into the grid to raise the voltage. Reference [26] shows configurations such as watt priority, which maximizes active power generation when the inverter operates at its power limit, while VV control requires additional reactive power to maintain voltage levels. The reactive power limits specified for VV control are given in (3).

$$Q_{pv,n} = \pm \sqrt{S_{pv,n}^2 - P_{pv,n}^2} \quad (3)$$

where $S_{pv,n}$ and $Q_{pv,n}$ are the n-th DER power rating and its available idle reactive power, respectively.

The behavior of VV control is shown in Fig. 2. As the voltage V changes in terms of per-units (pu), the reactive power follows the solid red line reference. Reactive power is injected when the voltage is below V_2 and absorbed when it exceeds V_4 . The voltage values between V_1 , V_2 , and V_4 , V_5 are calculated using a linearization method based on the reference points. The semicircle on the left shows the resultant power combination. For instance, for a voltage value between V_1 , V_2 , the resultant reactive power is set by reflecting it on the Y-axis, while maintaining the DER active power output with the watt priority configuration. If the inverter reaches its operational limits, active power takes precedence over reactive power, despite the inverter control reference.

The VW control aims to maintain acceptable voltage profiles by adjusting the active power output of a PV unit. Unlike VV control, VW control focuses solely on the upper voltage limit. When the voltage exceeds this threshold, the PV-based DER reduces injected active power, either to a predefined curtailment limit or down to zero. This approach is particularly effective in low X/R ratio grids with high DER generation, where overvoltage is a common issue. This paper employs a soft curtailment strategy, which limits active power

**FIGURE 2.** Illustrative volt-var curve.**FIGURE 3.** Illustrative Volt-Var and Volt-Watt curve.

reduction to the specific value, minimizing overall energy loss compared to hard curtailment [30], [46].

Fig. 3 shows the control behaviors for both VV and VW control strategies. In the VW control curve, P_2 is the maximum allowable curtailment, which is varied herein to analyze different levels of active power reduction. Regarding Fig. 3, the following operations are drawn:

- $V_{(pu)} < V_1$: DER supplies maximum active and reactive power;
- $V_1 < V_{(pu)} < V_2$: DER supplies maximum active power and provides a fraction of reactive power;
- $V_2 < V_{(pu)} < V_4$: DER supplies maximum active power;
- $V_4 < V_{(pu)} < V_5$: DER supplies maximum active power and absorbs a fraction of reactive power;
- $V_5 < V_{(pu)} < V_6$: DER supplies a fraction of active power and absorbs maximum reactive power;
- $V_{(pu)} > V_6$: DER supplies a fraction of active power and absorbs maximum reactive power.

The effectiveness of the strategies is evaluated by the activation frequency, which measures how often the control actions are triggered per PV unit.

III. PROPOSED TOOL FOR HOSTING CAPACITY ESTIMATION AND BESS SIZING

The HC value is estimated through several simulations with random allocations of PV units, and their rated power is increased step by step. The decentralized BESS sizing is

determined based on the estimated active power and energy curtailed by the VW control function embedded in the PV units.

A. BATTERY ENERGY STORAGE SIZING

In this paper, it is considered that the decentralized BESSs operate only during the curtailment period, in which VW control is activated, without a predetermined schedule. Equation (4) identifies the BESS maximum power capacity during the day. $BESS_{MaxPower}$ is the power that would be curtailed due to VW control implemented within the PV-based DER. $PV_{curtailed}[n]$ refers to the maximum amount of each n -th PV power curtailed at a specific time of the day. Equation (5) calculates the theoretical energy requirement based on power curtailed over time. $BESS_{Energy}$ is the useful energy capacity of the BESS, not accounting for the depth of discharge (DoD) and efficiency factors (Eff). $P_{curtailed}$ is the active power curtailed by VW control function, and $Duration$ is the time interval between samples. These factors are included in (6), where the DoD and battery efficiency are applied to compute the full energy capacity (i.e., $BESS_{Capacity}$). This paper uses kilowatt-hours (kWh) as metric instead of ampere-hours (Ah) and considers Li-ion battery technology.

$$BESS_{MaxPower}[\text{kW}] = \max\{PV_{curtailed}[n]\} \quad (4)$$

$$BESS_{Energy}[\text{kWh}] = \sum_0^N P_{curtailed}(n)[\text{kW}] \cdot Duration[\text{h}] \quad (5)$$

$$BESS_{Capacity}[\text{kWh}] = \frac{BESS_{Energy}[\text{kWh}]}{DoD[\%] \cdot Eff[\%]} \quad (6)$$

The DoD and Eff values are set to 0.80 and 0.98 respectively. They are applied to the useful $BESS_{Energy}$ to obtain the full $BESS_{Capacity}$ [47], [48]. It is performed for each BESS.

B. FIGURES OF MERITS

Four figures of merit are employed herein. The $E - \text{rate}$, (7), evaluates how quickly the battery is charged or discharged and can impact on the battery lifetime when operating at high rates [49]. Equation (8) is the ratio (i.e., $BessPVratio$) between the maximum battery power from (4) and the maximum power produced by the PV unit, PV_{power} . This ratio is a correlation between the BESS and the actual energy generation, underlining how effective the BESS can store the excess energy that would be curtailed by VW control.

Equation (9) evaluates the hosting capacity coefficient (HCC), which measures the energy curtailed by VW control relative to the increase in HC over the initial result without any control [50]. High values indicate that excessive curtailment occurs with minimal improvement in overall grid HC, which is unfavorable for investors because it results in substantial financial losses without significant benefits. Conversely, small values may show slight power reduction, suggesting that the VW control was inefficient. However,

they might also reflect a significant enhancement in HC due to minimal curtailment.

Equation (10) compares the output energy of the PV unit with the consumer demand (i.e., $PVLoadRatio$), offering insight into the power supply ratio and indicating how much of the demand is met by DER throughout the day. S_{PV} is the total energy generated by PV units over the day, while S_L is the total energy consumed by the equivalent load during the same period. However, it is important to consider the mismatch between the peak production of PV units, which generally occurs in the midday, and the distribution grid's load profile, which usually experiences peak demand from residential customers at the evening.

$$E - \text{rate} = \frac{BESS_{MaxPower}[\text{kW}]}{BESS_{Capacity}[\text{kWh}]} \quad (7)$$

$$BessPVratio = \frac{BESS_{MaxPower}[\text{kW}]}{PV_{Power}[\text{kW}]} \quad (8)$$

$$HCC(\%) = \frac{\text{Energy to be curtailed}}{\text{Energy above initial HC}} \quad (9)$$

$$PVLoadRatio(\%) = \frac{\sum S_{PV}}{\sum S_L} \quad (10)$$

C. TOOL FOR HOSTING CAPACITY ESTIMATION AND BESS SIZING

The proposed flowchart to evaluate the grid HC is shown in Fig. 4, which integrates three different platforms: OpenDSS, SQL Server, and Python, coloured in pink, blue, and yellow, respectively. The results were generated using OpenDSS version 9.6.1.2, SQL Server 2017, and Python 3.10. The py-DSS-interface and the Python-OpenDSS interface leverage OpenDSS functionalities without requiring its installation, as detailed in [51]. The Pandas library [52] is used for data manipulation. The communication between Python and the database is managed by the open-source SQLAlchemy library [53], an object-relational mapper that provides access to SQL functionalities via Python.

Before detailing the workflows, the simulation and case concepts are defined below, along with their scenarios:

- **Simulation:** is a unique randomly selected group of PV locations, which is evaluated for all proposed cases.
- **Case:** is a unique voltage control scenario associated with a simulation configuration under test.
 - **Case 1:** Simulates without PV units to establish the base grid operation.
 - **Case 2:** Simulates with PV units at unity power factor without control to establish baseline HC value.
 - **Case 3:** Simulates PV units with VV control to evaluate HC by controlling reactive power.
 - **Case 4:** Simulates PV units with VW control to evaluate HC by controlling active power.
 - **Case 5:** Simulates PV units with both VV and VW control to evaluate HC by managing both reactive and active power terms, with watt precedence.

The execution flow starts with the grid definition. The first pink square from Fig. 4 is the initial communication between Python and OpenDSS to set all elements of the grid. The process then check the current simulation count on the predefined total, proceeding if the count is lower. Otherwise, it triggers data processing. In summary, the process evaluates three soft curtailment levels for VW control — 15%, 30%, and 50% active power reduction — across 150 simulations for all case scenarios.

The simulation loops begin by defining PV installation locations, assuming a 20% PV penetration ratio based on the number of consumers, rounded up when needed. Once all configurations are set, the analysis starts with Case 1, which runs only once to collect grid operation data. In subsequent scenarios, PV generation starts at zero and increases in 1.5% increments of the rated transformer power up to a PQ violation occurs. If a violation is detected, the logic reduces the PV reference power by 1.5% and saves the result in the database as the HC value. This process repeats for all cases and simulations until completion. The final execution step is the data processing. Data is gathered from OpenDSS monitors, which collect various electrical parameters—such as voltage, current, and power — every 15 minutes for all grid elements. The electrical parameters as well as the HC values are stored in the database, while intermediate power flow data is discarded. The amount of stored data may vary by application, but each simulation generates about 1.5 million records across all case scenarios.

The data processing step involves calculations using existing data. The first data extracted includes PV active and reactive power, which are used to calculate HC values for each simulation case, following the procedure outlined in Section II-A. Based on the HC results, statistical analysis is conducted and stored in the database. The data required for sizing the BESS are also extracted and used following the procedure outlined in Section III-A.

IV. THE STUDIED LOW-VOLTAGE GRID

This section presents a case study based on a real LV grid in Américo Brasileiro, São Paulo, chosen for its validated HC limits in [17]. The methodology is generic, regardless of the voltage level, topology, or X/R characteristics of the evaluated grid, as long as proper attention is given to the power and voltage ratings of the practical converters. Fig. 5 shows the single-line grid diagram adopted herein, highlighting load and wire arrangements. The system includes daily load profiles for each consumer, where the demand varies dynamically throughout the day, representing a typical residential neighborhood with a radial configuration. The main branches are typically 3-phase, while most load connections are 1-phase, with some 2-phase and 3-phase loads. The grid includes 27 consumers, 41 buses, and 484.75 meters of cable, supplied by a 45 kVA transformer at the MV/LV coupling point. The transformer supplies a peak power of 14.408 kW and 4.777 kvar, with a daily energy demand of 160.876 kWh and 111.138 kvarh to meet the load requirements.

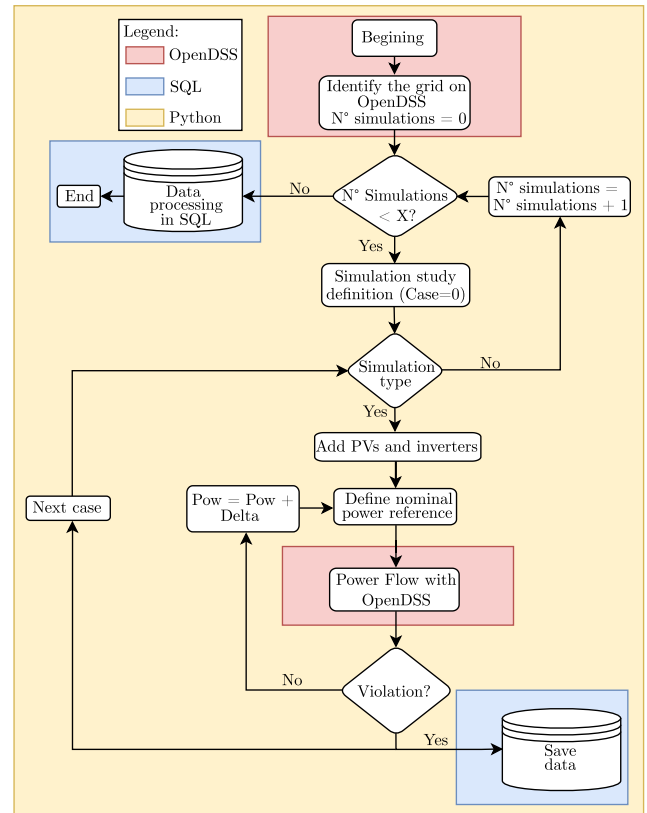


FIGURE 4. Flowchart of the developed tool using multiple software to estimate the HC value and size of the distributed small BESS units.

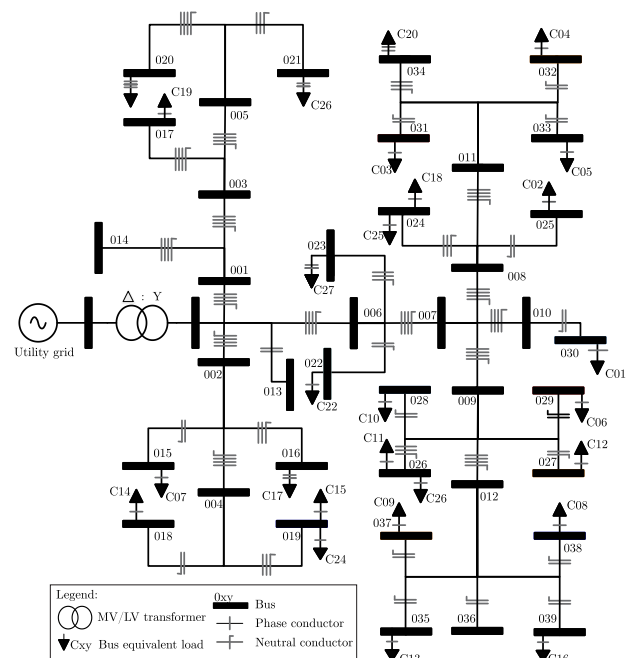


FIGURE 5. Low-voltage grid single-line diagram.

Table 4 compares the features between this case study and the [17], using similar parameters. The results in [17] were obtained from multiple LV grids, including the one used in

TABLE 4. Grid overall metrics comparison between the result obtained by [17] and herein.

Metric	Results from [17]	Case study
Circuit total length (m)	767.92	484.75
Average X/R ratio	1.01	0.72
Average conductor ampacity (A)	82.26	73.51
Average number of 1 ϕ costumers	20.41	21
Average number of 2 ϕ costumers	13.55	4
Average number of 3 ϕ costumers	2.77	2
Transformer rated power (kVA)	30 to 75	45
Average HC	37.40%	37.11%
HC standard deviation	17.40%	11.81%
HC	20%	25.3%

this case study (right column). Notably, the average circuit length is longer in [17], with a difference of 283.17 meters. The X/R ratio is slightly lower than [17], but remains within the comparable range considering the standard deviation of 0.32. This case study also has 70.47% fewer two-phase loads, although the number of single-phase and three-phase clients is nearly identical. The transformer-rated power in [17] results spans 30 kVA to 75 kVA, aligning with the 45 kVA transformer used herein.

Both studies assume that 20% of end-users have installed PV units. The authors of [17] provided a table of average HC values across different PV penetrations, based on data from multiple LV grids. Table 4 summarizes the results from both studies. The slight variations can be attributed to the different simulation methods used, as well as the number of iterations performed to obtain the average results.

V. SIMULATION RESULTS

The following sections discuss the enhancement of HC in a LV grid with a high penetration of rooftop PV inverters. The case study has a known HC value (i.e., 25.3%) when operating with PV units, ideal for comparing different solutions. Additionally, voltage control strategies, such as VV and VW control, are crucial for improving grid HC. While active power curtailment may reduce revenues when using VW control, the proposed decentralized BESS allows batteries to capture and store energy that would otherwise be lost.

A. HOSTING CAPACITY ANALYSIS

This section focuses on assessing HC based on the methodology of Section III. The same baseline serves as a reference for comparing the different levels of active power curtailment.

1) CURTAILMENT CONDITION AT 15%

The first scenario analyzed is the maximum 15% curtailment, with HC values shown in Table 5. The VV control strategy shows a significant improvement over the baseline, with a 37.07% increase compared to 1.38% improvement of VW control strategy. The highest enhancement is the combined VV and VW control, showing a 40.79% increase. The boxplot in Fig. 6(a) displays the data from Table 5.

TABLE 5. Average HC values for the proposed case scenarios with different levels of curtailment.

Curtailement	Cases	Average HC	Standard Deviation	HC	HC Enhancement
15%	Baseline	37.11%	11.81%	25.30%	-
	VV	49.46%	14.78%	34.68%	37.07%
	VW	39.16%	13.51%	25.65%	1.38%
	VV&VW	50.88%	15.26%	35.62%	40.79%
30%	Baseline	37.11%	11.81%	25.30%	-
	VV	51.76%	14.60%	37.16%	46.87%
	VW	42.45%	13.07%	29.38%	16.12%
	VV&VW	53.50%	14.77%	38.73%	53.08%
50%	Baseline	37.11%	11.81%	25.30%	-
	VV	50.99%	16.13%	34.86%	37.78%
	VW	47.01%	14.97%	32.04%	26.64%
	VV&VW	52.53%	16.50%	36.03%	42.41%

Compared to the baseline, the voltage control cases exhibit higher median and mean HC values, indicating that the grid can support more power from DERs. Additionally, the increase in the interquartile range (IQR) in the VV control and combined operations cases suggests a broader range of possible HC values.

2) CURTAILMENT CONDITION AT 30%

The second scenario analyzed is the maximum 30% curtailment, with the HC values shown in Table 5. The VV control resulted in a significant increase of 46.87% in HC, while the VW control experienced a 16.12% increase. The enhancement from VW control is greater than in the previous 15% curtailment scenario. The combined operation showed the most substantial improvement, with a 53.08% increase. The boxplot in Fig. 6(b) displays the data from Table 5. The voltage control cases show an increase in their median and mean HC values compared to the baseline results. The VW control case performed better under the 30% curtailment, achieving higher HC and IQR values than in the 15% scenario. Additionally, greater variability is evident in all control strategies, as indicated by the expanded IQR.

3) CURTAILMENT CONDITION AT 50%

The third scenario analyzed is the maximum 50% curtailment, with the HC values shown in Table 5. Similar to the previous scenarios, VV control shows a significant improvement of 37.78%. Meanwhile, VW control achieves a 26.64% increase, which is notably higher than in earlier scenarios. The combined operation achieved 42.41% enhancement. The boxplot in Fig. 6(c) displays the data from Table 5. Median and mean HC enhancements are evident, with VW control showing a particularly high response. Although the HC values across controls are quite similar, they address different metrics — active and reactive power. Additionally, the combined control operation results in a left-skewed distribution, indicating a higher likelihood of high HC values, as 25% of the samples cluster within a narrow quartile range.

4) CURTAILMENT CONDITION COMPARISON

Table 6 shows the enhancement of HC results, taking into account the following active power curtailment levels for VW

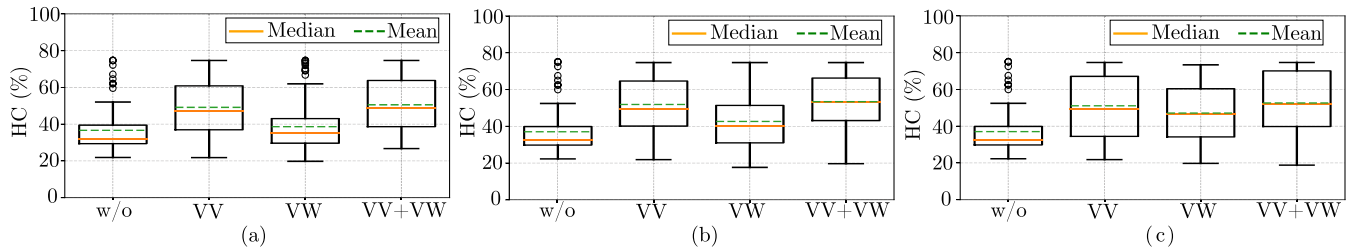


FIGURE 6. HC box plots: (a) 15% active power curtailment; (b) 30% active power curtailment; (c) 50% active power curtailment.

TABLE 6. Comparison of HC enhancements based on curtailment scenarios and voltage control techniques.

Control	15%→30%	15%→50%	30%→50%
VW	+14.54%	+24.91%	+9.05%
VV+VW	+8.73%	+1.15%	-6.97%

control and combined for VV+VW control strategies: 15% to 30%, 15% to 50%, and 30% to 50%. The VW control case demonstrated the most significant enhancement compared to the combined control operation, achieving increases of 14.54%, 24.91%, and 9.05% when comparing the 15% and 30%, 15% and 50%, and 30% to 50% curtailment scenarios, respectively. VW control stands out under higher active power reduction limits. However, higher curtailment results in more energy being wasted, which penalty owners by reducing profits. This challenge can be addressed through soft curtailment operations, as discussed in the next section, along with the implementation of BESS units to store energy that would otherwise be lost during VW control activation.

B. CONTROL STRATEGIES

This section analyzes the control activation, emphasizing the associated power losses and the relationship with the frequency of control activation. It also underlines the sensitivity of the random distribution of PV units over the LV grid.

1) DETAILED CONTROL ACTIVATION

Fig. 7(a) shows the control behavior of a single PV unit during a typical day. The RMS voltage measured across phases sets when the voltage support is triggered. The right Y-axis indicates control activation states: 0 is inactive, 1/ – 1 is full activation, and 0.2/ – 0.2 is partial activation. The left Y-axis displays pu voltage to represent the reference voltage throughout the day. The dashed lines depict the VV control limits at 0.99 pu and 1.01 pu, while the solid line is the VW control limit at 1.03 pu. These limits define when the control strategies are activated. As shown in Fig. 7(a), both VV and VW control are configured during certain periods. However, VW control may not activate if VV control stabilizes the bus voltage. If the DER requires more reactive power than it can handle to regulate the bus voltage, the PV watt priority configuration takes effect. As a result, partial VV

control is activated, with the PV unit supplying less reactive power than the reference value.

2) CURTAILED ACTIVE POWER

Herein, the same PV location, load conditions, and period are considered for a fair comparison of effects of curtailment strategies on PV active power generation. Three cases are addressed: without voltage support, VW control alone, and combined VV+VW control.

Figs. 7(b)–(c) focus on up to 15% curtailment. Under VW control, the active power reduction is 14.55% at 11:00 am - see Fig. 7(b), while combined control reduces this curtailment to 6.86% - see Fig. 7(c). Figs. 7(d)–(e) explore up to 30% curtailment. VW control leads to a 29.16% reduction at 10:00 am - see Fig. 7(d), which decreases to 12.41% with the combined control - see Fig. 7(e). Figs. 7(f)–(g) illustrate up to 50% curtailment. VW control results in a 49.16% reduction at 10:00 am, whereas combined control reduces it significantly to 13.75% - see Figs. 7(f) and (g), respectively. The results show that combining VV+VW control leads to a considerably lower curtailment, which is crucial for DER owners.

3) ACTIVATION COUNTS

Table 7 summarizes the control activations for different curtailment limits, with rows showing control strategies and columns indicating the percentage of simulations. 150 simulations are performed for each scenario, all using 6 PV units. In some cases, no PV unit has enabled voltage control, yet violations occur, meaning the grid reached its HC limit without issues on PV buses. As shown in Table 7, the findings for the 15%, 30%, and 50% curtailment limits are consistent. The VV control strategy required reactive power control activation from 5 and 6 PV units, while VW control and combined operation required activation of 2 and 3 PV units across all levels of curtailment.

The VV control is active in almost all units, benefiting both the utility and DER owners by enhancing the HC without requiring additional investment. The VW control scenario faces curtailment issues that decrease the active power delivered to the grid. Installing a BESS mitigates this loss by storing excess energy; however, it incurs costs for placing storage elements near the PV units. Achieving the right balance between curtailment limits and BESS size

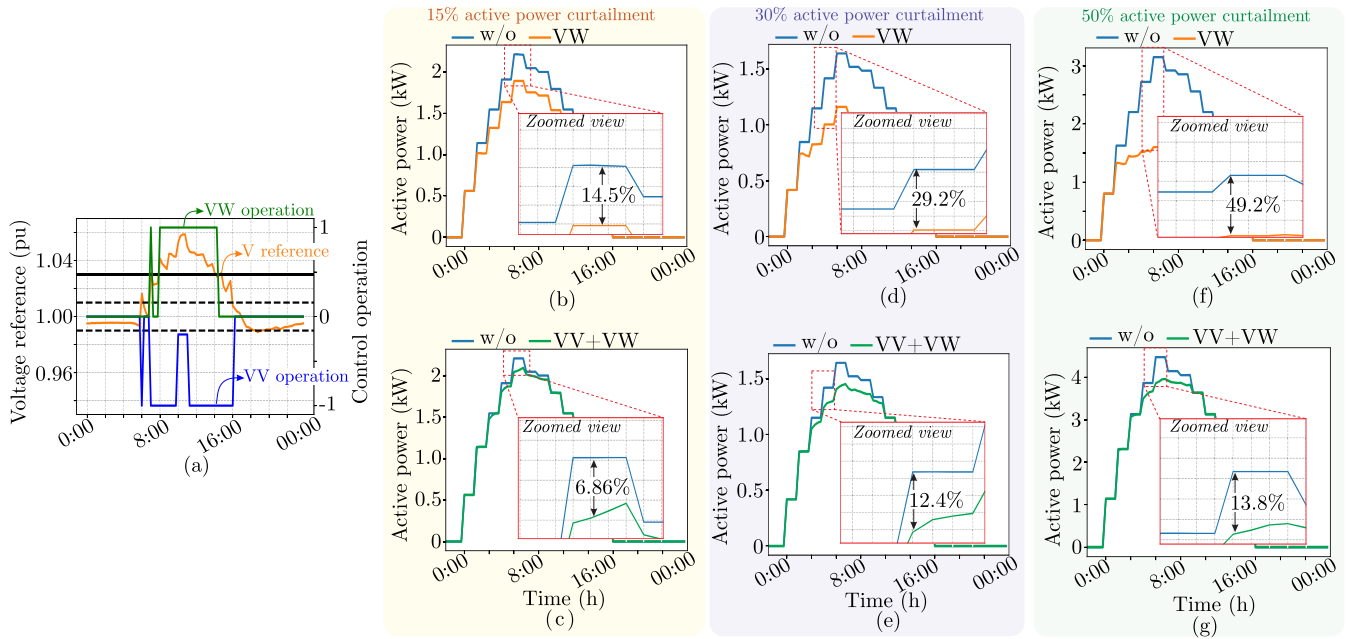


FIGURE 7. Voltage support strategies: (a) Control activation. PV-based DER considering up to 15% active power curtailment: (b) VW; (c) combined VV and VW control. PV-based DER considering up to 30% active power curtailment: (d) VW control; (e) combined VV and VW control. PV-based DER considering up to 50% active power curtailment: (f) VW control; (g) combined VV and VW control.

can enhance financial returns while maximizing the use of available physical space with more compact BESS units.

C. BESS SIZING ANALYSIS

This section discusses the sizing and performance indicators of BESS units designed to absorb power lost due to active power curtailment of PV units.

1) AVERAGE RESULTS OF BESS SIZING

Table 8 provides the average maximum power and energy requirements of BESS associated with each PV unit across different curtailment scenarios, including analyses for the VW control and the combined VW+VV control. It also includes the HC values to quantify the enhancement provided by decentralized BESS, according to the values shown in Table 5.

Considering the maximum 15% curtailment, the average maximum BESS power is 149.53 W (standard deviation: 70.97 W), and the average energy capacity is 859.26 Wh (standard deviation: 409.88 Wh). For maximum of 30% curtailment, these values rise to 338.96 W (standard deviation: 165.04 W) for power and 2022.16 Wh (standard deviation: 968.93 Wh) for energy capacity. Finally, for maximum of 50% curtailment, the requirements further increase to an average maximum active power of 729.24 W (standard deviation: 373.67 W) and an average energy capacity of 4527.96 Wh (standard deviation: 2429.22 Wh).

Additionally, similar findings for the combined VV+VW control scenario is observed in Table 8. For maximum of 15% curtailment limit, the average maximum operational power is 78.95 W (standard deviation: 43.69 W), and the

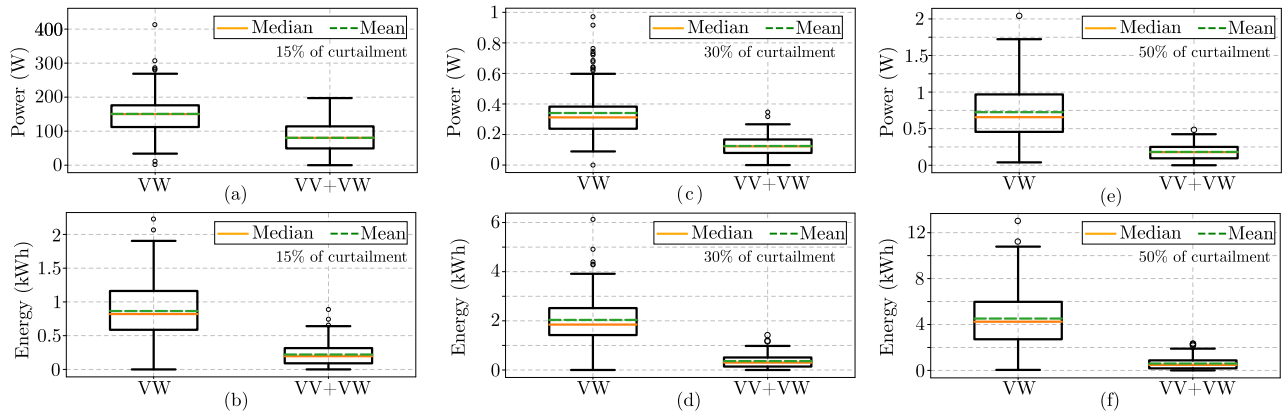
average energy capacity is 216.14 Wh (standard deviation: 168.17 Wh). For maximum of 30% curtailment, the average maximum active power is 126.39 W (standard deviation: 68.36 W), and the average energy capacity is 366.49 Wh (standard deviation: 284.31 Wh). Finally, for maximum of 50% curtailment, the average maximum active power is 183.32 W (standard deviation: 108.11 W), and the average energy capacity is 602.36 Wh (standard deviation: 511.36 Wh).

As expected, BESS maximum power and energy requirements increase as the allowable curtailment limit rises. However, the implementation of combined VV+VW control significantly reduces the size of the BESS units in both power and energy metrics. Specifically, the VW control shows an increase of 4.87 times when moving from 15% to 50% curtailment, while the VV+VW control scenarios exhibit a slighter increase of 2.32 times under the same conditions. There is a reduction in average BESS sizing when comparing the VW control with the combined VV+VW control scenarios, showing approximately 47.21% reduction for 15% curtailment, 62.71% reduction for 30%, and 74.86% reduction for 50% curtailment. These results account for all energy losses across the PV units considered in the simulation.

Fig. 8 shows the box plot results of BESS sizing (i.e., maximum battery power and battery adjusted energy) for 15%, 30%, and 50% active curtailment. Figs. 8(a), (c), and (e) show a notable reduction in both the average and median battery power for 15%, 30%, and 50% curtailment limit, respectively, when comparing the combined VW+VV control with VW control alone. The narrower whisker range suggests

TABLE 7. Number of elements operating in a HC condition with different voltage control strategies and different levels of curtailment.

Curtailment	Control function	Number of PV units that occurred activations of control functions							Simulations
		0	1	2	3	4	5	6	
15%	VV	4.6%	0.0%	0.6%	2.6%	12.6%	31.3%	48.0%	150
	VW	6.0%	9.3%	26.0%	32.6%	18.0%	7.3%	0.6%	150
	VV+VW	9.3%	10.6%	34.0%	28.0%	14.0%	4.0%	0.0%	150
30%	VV	0.0%	0.0%	0.0%	3.3%	14.0%	28.6%	54.0%	150
	VW	1.3%	10.6%	33.3%	32.0%	18.6%	4.0%	0.0%	150
	VV+VW	3.3%	16.0%	38.0%	34.6%	6.0%	2.0%	0.0%	150
50%	VV	0.0%	0.0%	0.0%	2.6%	10.0%	37.3%	50.0%	150
	VW	0.0%	5.3%	26.0%	29.3%	31.3%	8.0%	0.0%	150
	VV+VW	3.3%	18.0%	32.0%	32.6%	11.3%	2.6%	0.0%	150

**FIGURE 8.** Box plot results of BESS sizing: Maximum (a) power and (b) energy of battery bank considering 15% active curtailment. Maximum (c) power and (d) energy of battery bank considering 30% active curtailment. Maximum (e) power and (f) energy of battery bank considering 50% active curtailment.**TABLE 8.** Average results of BESS sizing by curtailment configuration for VW control and VV+VW control.

Control function	Case	Average HC	Average Maximum Power (W)	Standard Deviation	Battery Capacity (Wh)	Standard Deviation
VW	15%	25.65%	149.53	70.97	859.26	409.88
	30%	29.38%	338.96	165.04	2022.16	968.93
	50%	32.04%	729.24	373.67	4527.96	2429.22
VV + VW	15%	35.62%	78.95	43.69	216.14	168.17
	30%	38.73%	126.39	68.36	366.49	284.31
	50%	36.03%	183.32	108.11	602.36	511.36

that these reductions are concentrated toward smaller battery power values. Similar trends are observed in Figs. 8(b), (d), and (f) for the adjusted battery energy for 15%, 30%, and 50% curtailment limit, respectively.

The combined voltage support control quite reduces the size of the BESS. Consequently, the financial investment required for BESS installation is also diminished, as smaller BESS units are less costly and require less physical space.

2) PERFORMANCE INDICATORS FOR AVERAGE RESULTS OF BESS SIZING

Table 9 shows the average BESS sizing results for both VW control and combined VV+VW control operations, based on E-rate and *BessPVRatio* indicators as outlined in Section III-A. As shown in Table 9, the *BessPVRatio* increased by 229.01% and 422.85% when comparing the

TABLE 9. Average results of E-rate and PV power ratio by curtailment for VW control and VV+VW control.

Control	Case	HC	Average E-rate	Standard Deviation	Average BessPVRatio	Standard Deviation
VW	15%	25.65%	13.33%	5.23%	13.65%	4.88%
	30%	29.38%	30.70%	13.13%	31.26%	12.62%
	50%	32.04%	54.36%	37.36%	57.72%	35.97%
VV+VW	15%	35.62%	4.35	2.74	4.79%	2.56%
	30%	38.73%	6.43%	4.86%	7.14%	4.68%
	50%	36.03%	9.60%	7.27%	10.16%	7.13%

TABLE 10. Average results of PV by Load ratio energy by curtailment.

Curtailment	Baseline	VV	VW	VV&VW
15%	62%	90%	67%	94%
30%	62%	93%	74%	98%
50%	62%	91%	80%	96%

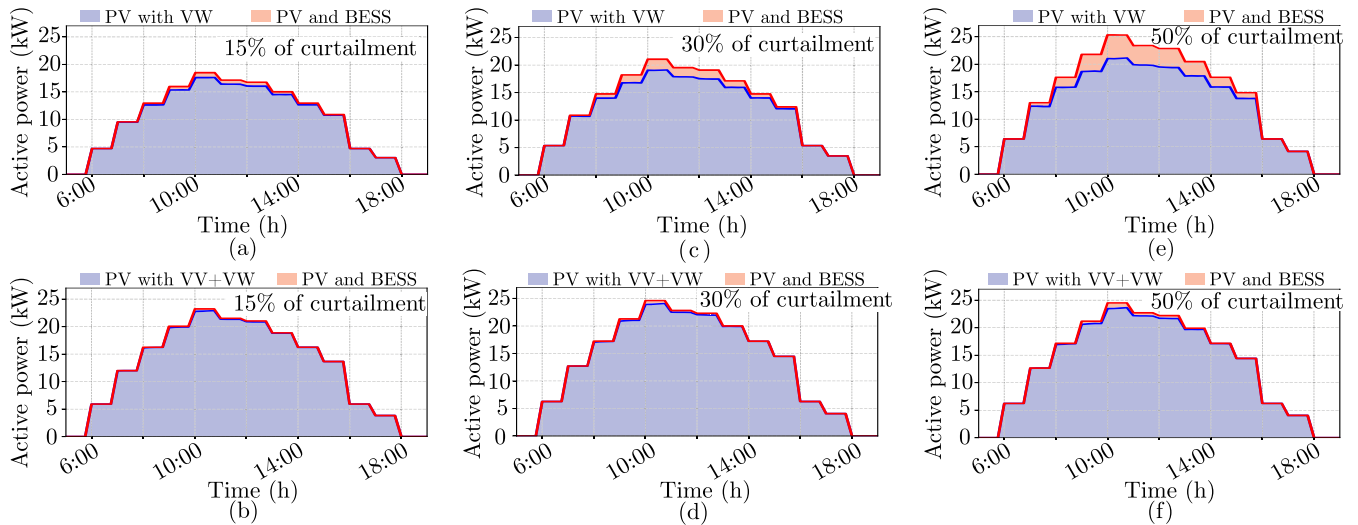
TABLE 11. Average results of HCC by curtailment configurations.

Case	VW Control			VV and VW Control		
	HC (%)	Average HCC (%)	Standard Deviation (%)	HC (%)	Average HCC (%)	Standard Deviation (%)
15%	25.65	12.80	136.48	35.62	2.48	23.82
30%	29.38	74.21	262.51	38.73	4.90	27.50
50%	32.04	132.12	351.91	36.03	16.56	30.29

15% curtailment with 30% and 50% for the VW control. In the combined VV+VW control, the increases are 149.06%

TABLE 12. Comparison results of average HC values considering PV with VW and VW control and BESS units for different curtailment.

Control function	Curtailment	With PV			With PV and BESS		
		Average HC (%)	Standard Deviation	HC (%)	Average HC (%)	Standard Deviation	HC (%)
VW	15%	39.16	13.51	25.65	41.08	13.66	27.42
	30%	42.45	13.07	29.38	46.81	13.17	33.64
	50%	47.01	14.97	32.04	56.25	15.82	40.43
VV+VW	15%	50.88	15.26	35.62	51.57	15.26	36.31
	30%	53.50	14.77	38.73	54.71	14.75	39.96
	50%	52.53	16.50	36.03	54.44	16.26	38.18

**FIGURE 9.** (a), (c), and (e): Power comparison from PV units equipped with VW control, with and without BESS, for curtailment scenarios of 15%, 30%, and 50%, respectively. (b), (d), and (f): Power comparison from PV units equipped with combined VV+VW control, with and without BESS, for curtailment scenarios of 15%, 30%, and 50%, respectively.

and 212.1%. Overall, as the curtailment limit increases, the BESS power converges toward the PV unit size. From Table 9, the E-Rate classifies the BESS as an E/3 unit, meaning it fully charges or discharges in 3 hours. This slow to medium charging rate effectively absorbs the power curtailed by the VW control and helps extend the lifetime of BESS units. Minimal differences are observed between control strategies, as both battery capacity and maximum power decrease proportionally, resulting in a consistent ratio across all configurations.

Table 10 shows the overall results for the *PVLoadRatio* indicator, as defined in Equation 10. The values compute the percentage of energy generated by all PV units relative to the daily load demand. There is a significant 46% increase in the *PVLoadRatio* for the VV control compared to the baseline. Under VW control, the ratio increases with higher curtailment levels, with a slight 1.08% rise at 15% curtailment and reaching 29.03% at 50%.

Table 11 shows the calculated HCC indicator, with average values and standard deviations for each control strategy and curtailment configuration. Both HC value and HCC indicator increase with higher curtailment levels as shown in Section III-A. Under VW control, the enhancements are 12.80% for the 15% curtailment, 74.21% for 30% curtailment, and 132.12% for 50% curtailment. For the

combined VV and VW control, the enhancements are 2.48%, 4.90%, and 16.56%, respectively. The smaller HCC in the combined control benefits both DER owners and utility, by producing HC enhancements with less curtailed energy and power loss.

3) HC RESULTS CONSIDERING DESCENTRALIZED BESS UNITS

The evaluation herein examines 150 different PV distributions for each curtailment level, showing the HC values as the average minus the standard deviation to provide a conservative analysis. The energy stored into BESS is considered in the numerator of HC calculation, as in (1). Table 12 shows the results for the PV unit operating under VW control and combined VV+VW control, referred to as cases 4 and 5, respectively. HC values increased across all curtailment levels for the VW control method, with growths of 6.9% for 15%, 14.49% for 30%, and 26.18% for 50% curtailment. Notably, for the 50% curtailment, HC increased from 32.04% to 40.43%, which aligns with the higher BESS power requirement shown in Table 8.

The HC enhancement is notably smaller for the combined VV+VW control compared to the VW control method, with increases of 1.93% for the 15% curtailment, 3.17% for the 30%, and 5.96% for the 50% curtailment levels. Overall,

HC enhancement remains within 6% across all curtailment levels, indicating a slight active power reduction in this case. It complies with the BESS size values in Table 8, inferring a reduced storage when combined control is employed in PV units.

The results from Table 12 are illustrated in Fig. 9, which shows the average daily three-phase power curves from all simulations. Figs. 9(a), (c), and (e) show the results for PV units equipped with VW control, with and without BESS, for curtailment scenarios of 15%, 30%, and 50%, respectively. Figs. 9(b), (d), and (f) show the results for PV units with VV+VW control under the same curtailment scenarios, also with and without BESS.

Fig. 9(a) shows a minor difference between scenarios with and without BESS, where maximum power decreases from 18.53 kW with BESS to 17.65 kW without. In the 30% curtailment scenario, Fig. 9(c) shows a more pronounced reduction, with maximum power dropping from 21.10 kW with BESS to 19.05 kW without. The most significant difference occurs in the 50% curtailment scenario - see Fig. 9(e), where maximum power declines from 25.33 kW with BESS to 21.04 kW without. It is worth noting that the increase in BESS storage energy, indicated by the red area, is not the sole factor contributing to higher HC values; the power injected by the PV units also increased.

A slight increase in the red area is observed in Figs. 9(b), (d), and (f) with BESS compared to the case without BESS. However, this increase is much smaller than that in Figs. 9(a), (c), and (e). For instance, the maximum power difference decreases from 23.29 kW with BESS to 22.76 kW without in Fig. 9(b). In Fig. 9(d), it changes from 24.57 kW with BESS to 23.93 kW without. In Fig. 9(f), the maximum power difference goes from 24.52 kW with BESS to 23.49 kW without. While these power differences are quite small, the average HC values remain consistently higher across all curtailment configurations compared to the results in Figs. 9(a), (c), and (e). The curtailed power has decreased. These results comply with the comparison between Table 12, which shows a smaller HC enhancement for the combined VV+VW control scenario. This finding is consistent with the BESS sizing analysis in Subsection V-C, where it was determined that VV+VW control requires smaller BESS unit.

VI. CONCLUSION

This paper quantified the use of decentralized BESS to mitigate overvoltage from high PV penetration in LV grids. A simulation tool was developed and proposed using OpenDSS, Python, and SQL to estimate the HC value. The case study evaluated 150 different PV allocations at 20% PV penetration based on the number of consumers. The results showed that autonomous VV control enhanced HC by approximately 1.35 times. VW control with curtailment limits of 15%, 30%, and 50% resulted in HC improvements of 1.02, 1.16, and 1.26 times, respectively, compared to the baseline scenario (i.e., HC = 25.3%). When combined (VV+VW

control), these strategies achieved improvements ranging from 1.40 to 1.53 times. These findings demonstrated the effectiveness of voltage support strategies and provide a quantification of HC gains when PV systems operate under these control schemes.

The power/energy of the BESS unit varied with the curtailment level, ranging from 13% to 57% for VW control and 4.79% to 10% for the combined VV+VW control, relative to the power/energy of the associated PV unit. When accounting for the power absorbed by the BESS in the HC calculation, the larger storage capacity required for VW control resulted in HC increases of 1.08, 1.33, and 1.60 times for 15%, 30%, and 50% curtailment scenarios, respectively, compared to the baseline. In contrast, the combined VV+VW control, which requires a smaller storage capacity, produced HC enhancements of 1.43, 1.58, and 1.51 times for 15%, 30%, and 50% curtailment levels.

The proposed tool effectively identifies HC values while incorporating voltage control techniques at PV units, and guides BESS units sizing to store energy that would otherwise be lost due to curtailment. Moreover, it is also capable of identifying the most sensitive buses of a LV grid. Therefore, results support the conclusion that the combined VV and VW control is the most suitable solution, as less storage capacity is required than VW control, while achieving higher HC. In terms of curtailment level and power/energy in decentralized BESS units, higher values result in greater HC enhancement. Finally, a promising direction for future research is to enhance the proposed methodology by incorporating additional decision-making metrics for BESS power rating selection, such as economic feasibility aspects [54]. Other potential extensions include selection of PV curtailment thresholds using machine learning or other techniques and expanding the analysis to grids at different voltage levels.

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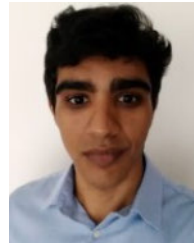
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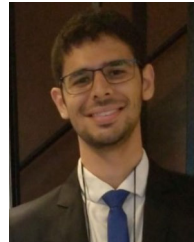
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